PRESSURE TRANSMITTER
PNR: 1-879-SB
MANUAL with Illustrated Parts List
ATA No

HIGHLIGHTS

Pages which have been added or revised are outlined below with the highlights of the revision. Remove and insert the pages concerned as listed and enter this revision.

REVISION No. 01 DATED JUL 27/26

Chapter/Section and Page No	Designation	Description of Change
Title Page TP 1 to TP 2	R	New presentation and new date.
Record of Revisions ROR Page 1 to ROR Page 2	N	Record of revision section added.
List of Effective Pages LEP Page 1 to LEP Page 2	N	List of effective pages section added.
Table of Contents TC Page 1 to TC Page 2	R	Revised to reflect this revision.
Introduction INTRO Page 1 to INTRO Page 2	N	Introduction section and shop verification added.
Repair Page 401	R	Environmental conditions updated.
Testing Page 701 to 704	R	Environmental conditions and minor technical corrections updated.
Storage Pages 901 to 902	R	Environmental conditions updated and desiccant bag deleted.

N: New - D: Deleted - R: Revised - U: Unchanged

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OVERHAUL MANUAL

First issue:

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79-30-02
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RECORD OF REVISIONS

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INTRODUCTION

1. General

This publication, prepared in accordance with ATA specification No 100 describes the functional and technical characteristics of PRESSURE TRANSMITTER and gives instructions for its maintenance.

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REPAIR

1. General

Faulty parts should be replaced by approved spares only. After any component has been renewed or repaired, carry out the full procedure in TESTING.

NOTE: All operations not described in this document must be done by the manufacturer, on their premises and in accordance with the approved industrial file.

NOTE: During repair operations, make sure that the environmental conditions are as follows:
- Pressure: No specific environmental level other than the ambient pressure level is necessary.
- Relative humidity: Refer to Standard Practices Manual ATA No. 20-00-02, F9111CMON-A-20-02-00-00A-800A-A - Storage - General procedure.

2. Procedure (Refer to Fig. 1101)

A. Bellow studs

(1) Damaged bellows stud may be removed in the following manner:

(a) Use a soldering iron to soften the solder which locks the stud to the bellows cover.

(b) Use a pair of flat-noised pliers to grip the stud then turn the stud counter-clockwise until it is clear of the bellows cover.

(c) Clean the bellows cover to remove all traces of solder and swarf etc.

(d) Select a new stud (Part No. 1872. AA. 01. 07)

(e) Screw the stud fully home into its hole in the bellows cover.

(f) Lock the stud to the bellows cover with solder.

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TESTING

1. General

The following tests are those which must be carried out after overhaul or whenever any component has been renewed or repaired. They are also the only tests necessary to establish the serviceability of a pressure transmitter.

NOTE: Before the tests, let the equipment stay in these conditions for two hours minimum.

NOTE: During testing operations, make sure that the environmental conditions are as follows:
- Temperature: 15 to 35 deg. C.
- Pressure: No specific environmental level other than the ambient pressure level is necessary.
- Relative humidity: Refer to Standard Practices Manual ATA No. 20-02-00, DMC,F9111CMON-A-20-02-00-00A-800A-A - Storage - General procedure.

2. Procedure

A. Visual Inspection: Refer to SPECIFICATIONS, page 5.

B. Checking Weight: 400 g ± 3% (0,882 lb ± 3%)

C. Accuracy Test: (Use one of the following two methods)

(1) With Test Stand 1.1045 CD

(a) Mount the transmitter in the test stand 1.1045 CD. The arrangement shown in fig. 1005 is suitable.

(b) Make the electrical connections between the transmitter, supply and master indicator as shown in fig. 507.

(c) With zero pressure applied to the transmitter, switch on the electrical supply. The master indicator should indicate zero within the tolerance band.

(d) Gradually apply pressure to the transmitter and take a series of readings in accordance with the following table:

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Table 701

Pressure applied (bars)	Pressure applied (p.s.i)	Tolerance (bars)	Tolerance (p.s.i)
0	0	± 0.15	± 2,17
1	14.50	± 0.15	± 2.17
2	29	± 0.15	± 2.17
3	43.50	± 0.15	± 2.17
4	58.00	± 0.15	± 2.17
5	72,50	± 0.15	± 2.17
6	87.00	± 0.15	± 2.17

(e) Decrease the pressure and re-check the reading of the master indicator at each of the given readings.

(f) The readings must be within the limits given in the above table. Should the transmitter fail this test, it must be re-calibrated.

(2) With Portable Test Set 1.1040 AF

(a) Connect the transmitter to a pressure line complete with a master pressure gauge calibrated from 0 to 100 p.s.i.

(b) Connect the transmitter to the Test Set as shown in fig. 701.

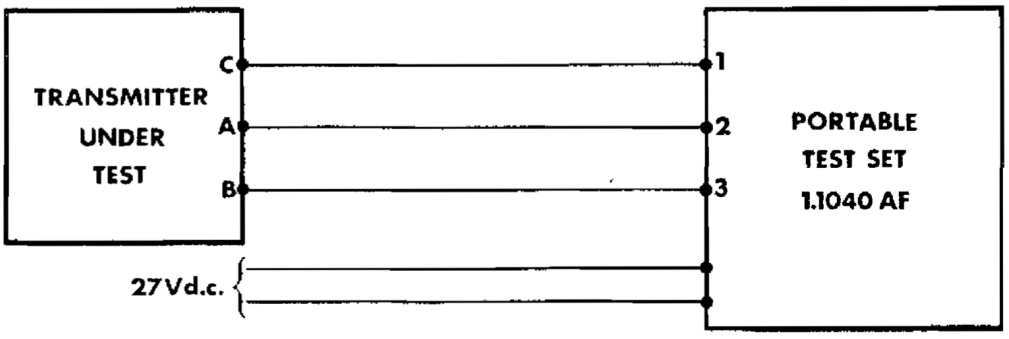


Figure 701

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(c) Refer to fig. 1004 and connect the Test Set to the power supply.

(d) On the Test Set, set the selector (15) to the left-hand 1.2 marking. Set the potentiometer knob (14) to 0 and switch on the electrical supply.

(e) With zero pressure applied to the transmitter, set the two resistance values, in turn, given in the following table, using the resistance box selectors (16, 17 and 18). At each of the two resistance values, press the push-button (10) and note the reading on the dial of the galvanometer (13). If the reading is not zero, bring it to zero by operating the potentiometer knob (14). When the galvanometer is reading zero, note the setting of the potentiometer knob.

(f) Gradually apply pressure to the transmitter. For each of the pressure values given in the following table, proceed as above, noting the setting of the potentiometer knob required to bring the reading of the galvanometer to zero.

(g) Decrease the applied pressure and re-note the setting of the potentiometer knob at each of the given readings to zero inclusive.

Table 702

Pressure applied (bars)	Pressure applied (p.s.i)	R 1 (ohms)	R 2 (ohms)
0	0	22.5	99.5
1	14.50	35.3	86.7
2	29	48.1	73.9
3	43.50	61	61
4	58.00	73.9	48.1
5	72.50	86.7	35.3
6	87.00	99.5	22.5

(h) The setting of the potentiometer knob required to null the galvanometer readings must not exceed $\pm$ 2 ohms. Should the transmitter fail this test, it must be re-calibrated.

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D. Lamp Operation Test

(1) With the transmitter connected as for the accuracy test, ensure that the warning contact operates at the correct pressures by carrying out the following procedure:

(a) Switch on the 27 V d.c. supply and observe that, with zero pressure applied to the transmitter, the test warning lamp lights.

(b) Apply a pressure of 6 bars (87 p.s.i.) to the transmitter.

(c) Gradually reduce the pressure and observe that the test lamp lights at 0.7 bar $\pm$ 50 millibars ($\pm$ 0.7 p.s.i.) and remains illuminated when the pressure is reduced to zero.

E. Overload Test

(1) With the transmitter still connected as for the accuracy test, but with the power supply switched off, apply a pressure of 7.5 bars (108 p.s.i.) for a period of two minutes.

(2) Allow a recovery time of 24 hours and repeat the accuracy test. There must be no reduction in accuracy.

F. Insulation Resistance Test

(1) Use a 500 V d.c. insulation tester to measure the resistance between all terminal pins joined together and earth (cover).

(2) The insulation resistance value must not be less than 40 megohms.

G. Dielectric Breakdown Test

(1) Connect a flash tester between the terminal pins connected together and earth. Apply 500 V d.c. for a period of one minute.

(2) Check that there is no tracking or insulation breakdown, as would be indicated by a drop in the meter reading.

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STORAGE

1. General

NOTE: During storage operations, make sure that the environmental conditions are as follows:
- Pressure: No specific environmental level other than the ambient pressure level is necessary.
- Relative humidity: Refer to Standard Practices Manual ATA No. 20-02-00, F9111CMON-A-20-02-00-00A-800A-A - Storage - General procedure.

A. Short Storage Period

If the storage period is not more than two months under normal atmospheric conditions, the instrument should be packed in the following manner:

(1) Wrap the instrument in kraft paper.

(2) Insert the instrument in a polyvinyl bag.

(3) Draw the vacuum and heat-seal the bag.

(4) Fit the bag into a cardboard box cushioned with a shock-absorbent material.

(5) Close and seal the box with adhesive tape.

(6) Mark the package with a label stuck on to the top.

B. Long Storage Period

When the storage period is extended or the instrument is to be preservated from severe atmospheric conditions, the package is inserted into a reserviceable metal container. The instrument should be packed in the following manner:

(1) Wrap the instrument in a greaseproof paper (greaseproof side toward the instrument).

(2) Seal the bag by sticking (the greaseproof paper is adhesive).

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(3) Fit the bag into a metal container.

(4) Consolidate the instrument in the container using deshydrated cushioning materials.

(5) Enclose an humidity indicator or an indicator card.

(6) Seal the container.

(7) Mark the package by stenciling on the side of the container the following information: Nomenclature, serial number, packaging date.

C. Unpacking

After the recommended period in storage and before its installation, the instrument must be unpacked and tested in accordance with the procedure described in TESTING.

Test:
- do (1),
- do (2),
- do (3)
- do (4)
- put a dessicant bag,
- do (5).

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